

Café Sales Analysis: Turning Transaction Data into Actionable Business Insights

(An independent portfolio case study applying skills developed through the Google Data Analytics Professional Certificate)

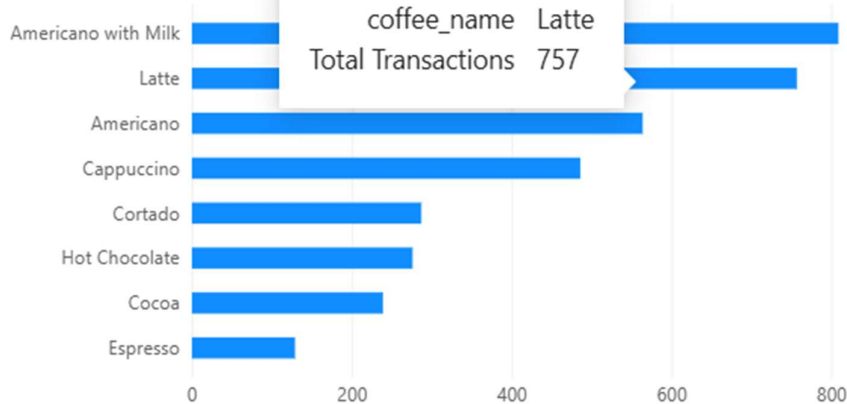
Tools: Excel, SQL(PostgreSQL), Power BI, DAX

Dataset: Café Sales Transaction Data (Kaggle)

Analysis Period: March 2024- March 2025

Prepared by: Harshada Atpadkar

Total Transactions
by coffee_name



3547	112.25K
Total Transactions	Total Transaction Value

31.65	8
Average Transaction Value	Number of Products

Project Overview:

Café transaction data contains information about products, transaction values, purchase times & dates, but raw transaction records do not directly reveal which products perform best, when transaction activity peaks or how demand varies across weekdays and time periods. This case study analyses café transaction data to identify product performance, purchasing patterns, peak transaction periods, weekday demand and changes in transaction value over time, that can support operational and commercial decision-making.

Business Objectives:

The case study was designed to answer the following questions:

Product Performance

- 1. Which products generate the highest transaction value?
- 2. Which products have the highest transaction volume?

Customer Demand

- 3. What are the peak hours for cafe transactions?
- 4. Which weekdays experience the highest and lowest transaction activity?

Sales Patterns

- 5. How does transaction value vary across different times of day?
- 6. How does transaction value change over the analysis period?



Figure 1. Analytical objectives

Dataset Overview:

Table 1. Dataset Overview

Metric	Value
Transactions	3,547
Products	8
Analysis Period	March 2024 – March 2025
Total Transaction Value	112.25K
Average Transaction Value	31.65

Data Preparation:

- Reviewed the dataset structure and variables

Table 2. Key Dataset Variables and Analytical Purpose

Variable	Purpose
coffee_name	Product performance
money	Transaction value
hour_of_day	Peak-hour analysis
Time_of_Day	Time-period analysis
Weekday	Weekday demand
Date	Trend analysis
cash_type	Payment-method analysis

- Checked and converted date and time fields in proper format, for analysis
- Prepared weekday and month fields for time-based analysis
- Created sorting fields to maintain chronological and weekday order
- Performed exploratory analysis using SQL (PostgreSQL)
- Prepared calculated measures for transaction & value analysis in Power BI
- Added interactive filters for product, time period and date analysis

	A	B	C	D	E	F	G	H	I	J	K	L
1	Transaction_ID	cash_type	money	coffee_name	Time_of_Day	Weekday	Month_name	Weekdaysort	Monthsort	Date	Time	hour_of_day
2	TXN-00001	card	38.7	Latte	Morning	Fri	Mar	5	3	01-03-2024	10:15:51	10
3	TXN-00002	card	38.7	Hot Chocolate	Afternoon	Fri	Mar	5	3	01-03-2024	12:19:23	12
4	TXN-00003	card	38.7	Hot Chocolate	Afternoon	Fri	Mar	5	3	01-03-2024	12:20:18	12
5	TXN-00004	card	28.9	Americano	Afternoon	Fri	Mar	5	3	01-03-2024	13:46:33	13
6	TXN-00005	card	38.7	Latte	Afternoon	Fri	Mar	5	3	01-03-2024	13:48:15	13
7	TXN-00006	card	33.8	Americano with Milk	Afternoon	Fri	Mar	5	3	01-03-2024	15:39:48	15
8	TXN-00007	card	38.7	Hot Chocolate	Afternoon	Fri	Mar	5	3	01-03-2024	16:19:03	16
9	TXN-00008	card	33.8	Americano with Milk	Night	Fri	Mar	5	3	01-03-2024	18:39:04	18
10	TXN-00009	card	38.7	Cocoa	Night	Fri	Mar	5	3	01-03-2024	19:22:02	19
11	TXN-00010	card	33.8	Americano with Milk	Night	Fri	Mar	5	3	01-03-2024	19:23:16	19
12	TXN-00011	card	33.8	Americano with Milk	Night	Fri	Mar	5	3	01-03-2024	19:29:17	19
13	TXN-00012	card	28.9	Americano	Morning	Sat	Mar	6	3	02-03-2024	10:22:07	10
14	TXN-00013	card	33.8	Americano with Milk	Morning	Sat	Mar	6	3	02-03-2024	10:41:41	10
15	TXN-00014	card	33.8	Americano with Milk	Morning	Sat	Mar	6	3	02-03-2024	11:59:45	11
16	TXN-00015	card	28.9	Americano	Afternoon	Sat	Mar	6	3	02-03-2024	14:38:36	14
17	TXN-00016	card	33.8	Americano with Milk	Afternoon	Sat	Mar	6	3	02-03-2024	16:37:24	16
18	TXN-00017	card	28.9	Americano	Night	Sat	Mar	6	3	02-03-2024	17:34:55	17
19	TXN-00018	card	38.7	Latte	Morning	Sun	Mar	7	3	03-03-2024	10:27:19	10
20	TXN-00019	card	28.9	Cortado	Morning	Sun	Mar	7	3	03-03-2024	11:33:56	11
21	TXN-00020	card	28.9	Americano	Afternoon	Sun	Mar	7	3	03-03-2024	12:26:56	12
22	TXN-00021	card	38.7	Hot Chocolate	Afternoon	Sun	Mar	7	3	03-03-2024	13:09:36	13
23	TXN-00022	card	38.7	Cocoa	Night	Sun	Mar	7	3	03-03-2024	17:06:40	17
24	TXN-00023	card	28.9	Cortado	Night	Sun	Mar	7	3	03-03-2024	17:08:46	17
25	TXN-00024	card	33.8	Americano with Milk	Night	Sun	Mar	7	3	03-03-2024	18:03:23	18

Figure 2. Cleaned Café Sales Data

Analytical distinction: Transaction volume and transaction value were analysed separately because purchase frequency and monetary contribution represent different aspects of product performance.

SQL Analysis:

Table 3. SQL Queries

1.Product Performance Ranking: Select coffee_name, count(*) as transactions, sum(money) as total_value, rank() over (order by sum(money) desc) as value_rank from cafe_sales group by coffee_name order by value_rank;	<table><tr><th></th><th>coffee_name character varying (100)</th><th>transactions bigint</th><th>total_value numeric</th><th>value_rank bigint</th></tr><tr><td>1</td><td>Latte</td><td>757</td><td>26875.30</td><td>1</td></tr><tr><td>2</td><td>Americano with Milk</td><td>809</td><td>24751.12</td><td>2</td></tr><tr><td>3</td><td>Cappuccino</td><td>486</td><td>17439.14</td><td>3</td></tr><tr><td>4</td><td>Americano</td><td>564</td><td>14650.26</td><td>4</td></tr><tr><td>5</td><td>Hot Chocolate</td><td>276</td><td>9933.46</td><td>5</td></tr><tr><td>6</td><td>Cocoa</td><td>239</td><td>8521.16</td><td>6</td></tr><tr><td>7</td><td>Cortado</td><td>287</td><td>7384.86</td><td>7</td></tr><tr><td>8</td><td>Espresso</td><td>129</td><td>2690.28</td><td>8</td></tr></table>		coffee_name character varying (100)	transactions bigint	total_value numeric	value_rank bigint	1	Latte	757	26875.30	1	2	Americano with Milk	809	24751.12	2	3	Cappuccino	486	17439.14	3	4	Americano	564	14650.26	4	5	Hot Chocolate	276	9933.46	5	6	Cocoa	239	8521.16	6	7	Cortado	287	7384.86	7	8	Espresso	129	2690.28	8	Result: Latte ranked the highest by total transaction value																														
	coffee_name character varying (100)	transactions bigint	total_value numeric	value_rank bigint																																																																									
1	Latte	757	26875.30	1																																																																									
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8	Espresso	129	2690.28	8																																																																									
2. Peak hour by Time of Day: Select time_of_day, hour_of_day, count(*) as transactions, rank() over (partition by time_of_day order by count(*) desc) as hour_rank from cafe_sales group by time_of_day, hour_of_day;	<table><tr><th></th><th>time_of_day character varying (20)</th><th>hour_of_day numeric (10,2)</th><th>transactions bigint</th><th>hour_rank bigint</th></tr><tr><td>1</td><td>Afternoon</td><td>16.00</td><td>278</td><td>1</td></tr><tr><td>2</td><td>Afternoon</td><td>12.00</td><td>241</td><td>2</td></tr><tr><td>3</td><td>Afternoon</td><td>15.00</td><td>236</td><td>3</td></tr><tr><td>4</td><td>Afternoon</td><td>14.00</td><td>225</td><td>4</td></tr><tr><td>5</td><td>Afternoon</td><td>13.00</td><td>225</td><td>4</td></tr><tr><td>6</td><td>Morning</td><td>10.00</td><td>328</td><td>1</td></tr><tr><td>7</td><td>Morning</td><td>11.00</td><td>283</td><td>2</td></tr><tr><td>8</td><td>Morning</td><td>9.00</td><td>242</td><td>3</td></tr><tr><td>9</td><td>Morning</td><td>8.00</td><td>235</td><td>4</td></tr><tr><td>10</td><td>Morning</td><td>7.00</td><td>88</td><td>5</td></tr><tr><td>11</td><td>Morning</td><td>6.00</td><td>5</td><td>6</td></tr><tr><td>12</td><td>Night</td><td>17.00</td><td>237</td><td>1</td></tr><tr><td>13</td><td>Night</td><td>19.00</td><td>229</td><td>2</td></tr><tr><td>14</td><td>Night</td><td>18.00</td><td>210</td><td>3</td></tr></table>		time_of_day character varying (20)	hour_of_day numeric (10,2)	transactions bigint	hour_rank bigint	1	Afternoon	16.00	278	1	2	Afternoon	12.00	241	2	3	Afternoon	15.00	236	3	4	Afternoon	14.00	225	4	5	Afternoon	13.00	225	4	6	Morning	10.00	328	1	7	Morning	11.00	283	2	8	Morning	9.00	242	3	9	Morning	8.00	235	4	10	Morning	7.00	88	5	11	Morning	6.00	5	6	12	Night	17.00	237	1	13	Night	19.00	229	2	14	Night	18.00	210	3	Result: Identifies the peak transaction hour within each time-of- day segment
	time_of_day character varying (20)	hour_of_day numeric (10,2)	transactions bigint	hour_rank bigint																																																																									
1	Afternoon	16.00	278	1																																																																									
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7	Morning	11.00	283	2																																																																									
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3. Weekday Performance: Select weekday, count(*) as transactions, round(100.0*count(*) /sum(count(*)) over (), 2) as share_pct from cafe_sales group by weekday order by transactions desc;	<table><tr><th></th><th>weekday character varying (20)</th><th>transactions bigint</th><th>share_pct numeric</th></tr><tr><td>1</td><td>Tue</td><td>572</td><td>16.13</td></tr><tr><td>2</td><td>Mon</td><td>544</td><td>15.34</td></tr><tr><td>3</td><td>Fri</td><td>532</td><td>15.00</td></tr><tr><td>4</td><td>Thu</td><td>510</td><td>14.38</td></tr><tr><td>5</td><td>Wed</td><td>500</td><td>14.10</td></tr><tr><td>6</td><td>Sat</td><td>470</td><td>13.25</td></tr><tr><td>7</td><td>Sun</td><td>419</td><td>11.81</td></tr></table>		weekday character varying (20)	transactions bigint	share_pct numeric	1	Tue	572	16.13	2	Mon	544	15.34	3	Fri	532	15.00	4	Thu	510	14.38	5	Wed	500	14.10	6	Sat	470	13.25	7	Sun	419	11.81	Result: Tuesday recorded the highest transaction volume with 572 transactions																																											
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1	Tue	572	16.13																																																																										
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6	Sat	470	13.25																																																																										
7	Sun	419	11.81																																																																										
4.Daily sales Trends: Select date, sum(money) as daily_value, avg(sum(money)) over (order by date rows between 6 preceding and current row) as moving_avg_7d from cafe_sales group by date order by date;	<table><tr><th></th><th>date date</th><th>daily_value numeric</th><th>moving_avg_7d numeric</th></tr><tr><td>1</td><td>2024-03-01</td><td>396.30</td><td>396.3000000000000000</td></tr><tr><td>2</td><td>2024-03-02</td><td>188.10</td><td>292.2000000000000000</td></tr><tr><td>3</td><td>2024-03-03</td><td>309.10</td><td>297.8333333333333333</td></tr><tr><td>4</td><td>2024-03-04</td><td>135.20</td><td>257.1750000000000000</td></tr><tr><td>5</td><td>2024-03-05</td><td>338.50</td><td>273.4400000000000000</td></tr><tr><td>6</td><td>2024-03-06</td><td>135.20</td><td>250.4000000000000000</td></tr><tr><td>7</td><td>2024-03-07</td><td>140.10</td><td>234.6428571428571429</td></tr><tr><td>8</td><td>2024-03-08</td><td>265.50</td><td>215.9571428571428571</td></tr></table>		date date	daily_value numeric	moving_avg_7d numeric	1	2024-03-01	396.30	396.3000000000000000	2	2024-03-02	188.10	292.2000000000000000	3	2024-03-03	309.10	297.8333333333333333	4	2024-03-04	135.20	257.1750000000000000	5	2024-03-05	338.50	273.4400000000000000	6	2024-03-06	135.20	250.4000000000000000	7	2024-03-07	140.10	234.6428571428571429	8	2024-03-08	265.50	215.9571428571428571	Result: The 7-day moving average helps identify the underlying sales trend while reducing daily fluctuations																																							
	date date	daily_value numeric	moving_avg_7d numeric																																																																										
1	2024-03-01	396.30	396.3000000000000000																																																																										
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Power BI Dashboard



Figure 3. Power BI Dashboard - Café Sales Analysis

The dashboard integrates key performance indicators with product, hourly, weekday, times-of-day and trend analysis. Interactive slicers allow users to explore patterns across products, dates and time periods.

Dashboard Components:

- **KPI Overview:** Transactions, transaction value, average transaction value and product count
- **Product Analysis:** Comparison of transaction volume and transaction value by coffee
- **Demand Analysis:** Hourly and weekday transaction patterns.
- **Time-based Analysis:** Transaction value across time of day and the analysis period.

Key Findings

1. Product Performance



Highest Transaction Volume

Americano with milk - 809

Recorded the highest transaction volume, indicating strong purchase frequency.

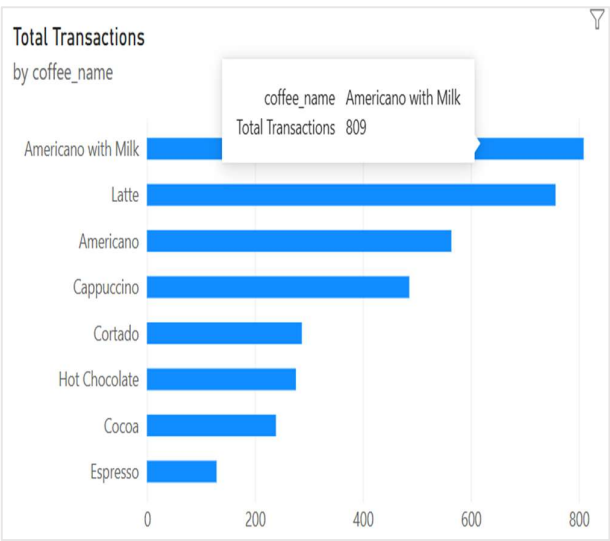


Figure 4. Transaction volume by coffee



Highest Transaction Value

Latte – 26,875.30

Generated the highest aggregate transaction value, showing the importance of evaluating both volume and monetary contribution.

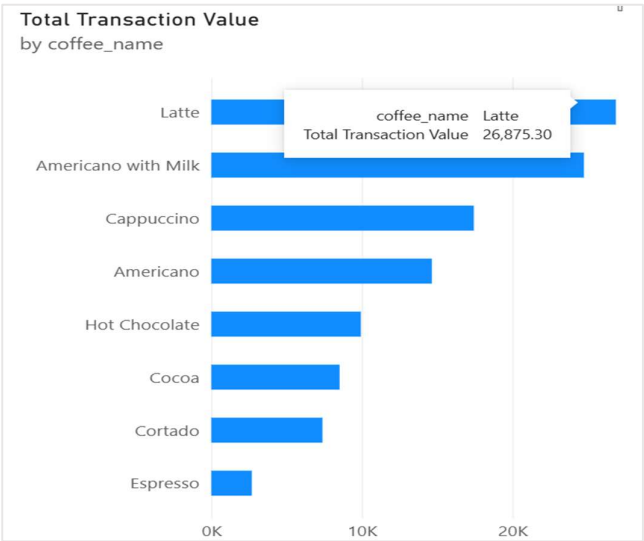


Figure 5. Transaction value by coffee

2. Demand Patterns



Peak Hours

10 AM – 328 transactions

Recorded the highest hourly transaction volume

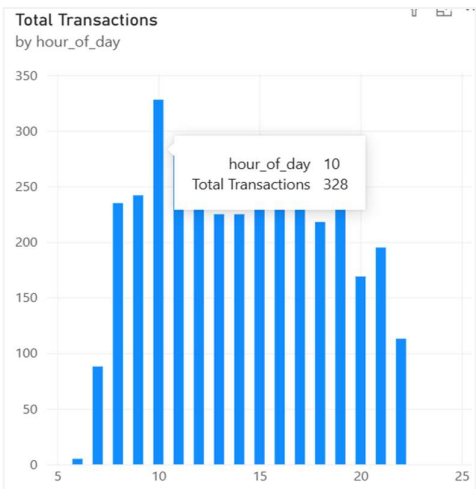


Figure 6. Transactions by Hour



Highest & Lowest Weekday

Tuesday – 572 & Sunday – 419 Tr.

Tuesday-the highest weekday transaction volume
Sunday- The lowest weekday transaction volume

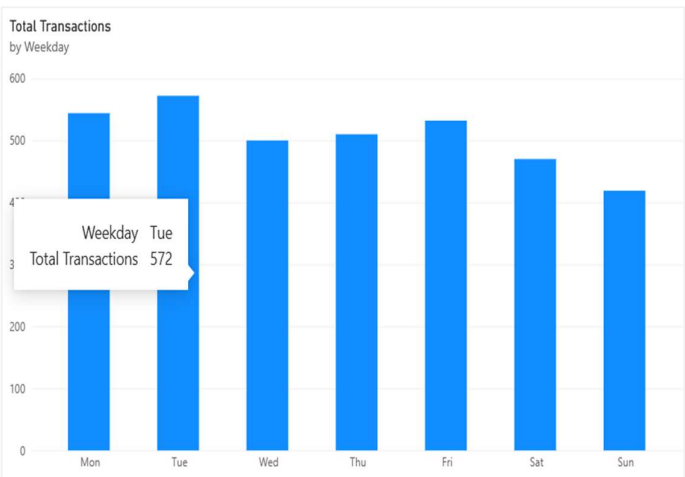


Figure 7. Transactions by weekday

3. Daily Transaction Value Trend



Highest Observed Date
2024 October 11 – 836.66



Lowest Observed Date
2024 August 29 – 23.02

Transaction value varied across the analysis period, with these dates representing the highest and lowest observed values

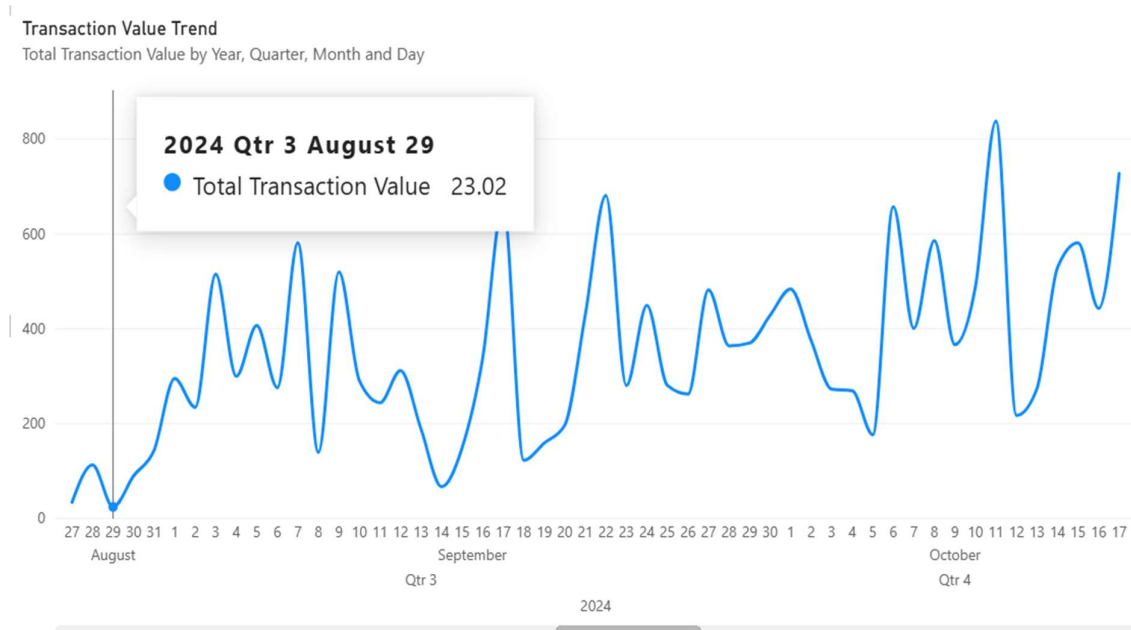


Figure 8. Daily Transaction Value Trend Over the Analysis Period

Business Recommendations:

The findings from the transaction, product and time-based analysis can be translated into several data-informed business actions. These recommendations focus on improving operational planning, product management and demand-based decision-making.

Table 4. Data-Informed Business Recommendations

Recommendation	Finding	Action
1. Inventory planning	'Americano with milk' recorded the highest transaction volume with 809 transactions.	Monitor inventory levels of high-volume products & align replenishment with observed demand patterns to reduce potential stock-outs.
2. Peak-hour staffing	10 AM recorded the highest hourly transaction volume with 328 transactions.	Consider aligning staffing & operational capacity around peak periods to manage customer demand efficiently.
3. Lower-volume day strategy	Sunday recorded the lowest weekday transaction volume with 419 transactions.	Consider targeted promotions, loyalty offers or other demand-building initiatives on relatively lower-volume days.
4. Product Portfolio Management	'Americano with milk' led transaction volume, while Latte led total transaction value.	Evaluate products using both volume & monetary contribution when making product portfolio decisions.
5. Continuous Monitoring	Transaction value varied across the analysis period.	Use the Power BI dashboard as a recurring monitoring tool to identify changes in product & demand patterns & support ongoing decision-making.

Limitations:

This analysis is based on historical transactional-level data and focuses primarily on transaction volume & transactional value. The dataset doesn't include customer demographics, customer-level purchase history, discounts, promotional campaigns, profit margins, weather conditions, holidays or external events. As a result, the analysis identifies patterns & relationships within the available data but doesn't establish the underlying causes of those patterns. The recommendations should therefore be viewed as data-informed opportunities for further investigation rather than definitive causal conclusions. Additional customer, operational and external-context data would enable deeper analysis & more precise business decision-making.

Conclusion:

This case study demonstrates how transactional-level café data can be transformed into meaningful business insights through structured data analysis, SQL and Power BI. These insights can support data-informed decisions related to inventory planning, staffing, product portfolio management, promotional strategies and ongoing performance monitoring.

Overall, the project strengthened my ability to move from raw data → analysis → visualization → insights → business recommendations, applying data analytics concepts in a practical business context.